

ONLINE RETAIL SALES & CUSTOMER ANALYTICS

Project Documentation

Tools used: Excel, MySQL and Power BI

Dataset: UCI Online Retail Dataset

Period covered: December 2010 – December 2011

This document explains the project in simple language. It shows what I looked at, how I worked with the data, what I found, and what the findings could mean for the business.

1. What This Project Is About

This project looks at sales data from an online retailer in the UK. The data contains information about orders, products, customers, countries, prices and cancellations.

I wanted to turn the raw data into information that a business owner or manager could easily understand and use.

The main questions were:

- How much money did the business make?
- Which products sold the most?
- Which countries brought in the most money?
- How often did customers come back and buy again?
- How large was the impact of cancelled orders?
- When were sales strongest?
- What could the business do better based on the results?

2. The Data

The project uses the UCI Online Retail Dataset. It contains sales records from a UK-based online retailer.

The original file contained 541,909 records. After checking the data and removing exact duplicate records, 536,641 records remained for the final analysis.

The data includes information such as:

- Invoice number
- Product code and product name
- Quantity bought
- Date and time of the transaction
- Price
- Customer number
- Country
- Cancellation information

3. How I Prepared the Data

Before doing any analysis, I checked the data carefully. The goal was to make sure the numbers used in the final report were reliable.

The main cleaning steps were:

- Removed exact duplicate records.
- Made the column names clear and consistent.
- Separated the date and time of each transaction.
- Added a field to identify cancelled transactions.
- Calculated the value of each transaction line using quantity × price.
- Added categories to help explain unusual records.

- Checked missing customer information.
- Kept unusual records when they had a valid business reason instead of simply deleting them.

For example, some records had negative quantities. Not all of these were customer cancellations. I therefore checked them before deciding how to treat them.

The final figures were checked in both Excel and MySQL so that the results could be compared and confirmed.

4. Main Results

The final analysis showed:

Measure	Result
Records analysed	536,641
Net quantity	5,162,502
Total revenue	£9,726,006.95
Customers identified	4,372
Countries served	38
Cancellation rate	1.72%
Cancellation value	£893,979.73

These figures give a quick picture of the size of the business during the period covered by the data.

5. What I Found About Customers

One of the strongest findings was the difference between customers who bought once and customers who came back to buy again.

Customer results

- About 65.6% of customers with positive purchases were repeat customers.
- Repeat customers generated 93.09% of positive-sales revenue linked to identified customers.
- Repeat customers generated about £2,908 each on average.
- One-time customers generated about £411 each on average.

This suggests that returning customers are very important to the business. Keeping existing customers may therefore be just as important as finding new ones.

6. What I Found About Products

I looked at products in two main ways: how much money they generated and how many units were sold.

The highest-revenue physical products included the Regency Cake Stand 3 Tier, White Hanging Heart T-Light Holder, Party Bunting and Jumbo Bag Red Retrospot.

I kept postage, fees and other financial entries separate from physical products when looking at product performance. This gives a clearer picture of which actual products were driving sales.

7. What I Found About Countries

The UK was by far the largest market.

- The UK generated 83.97% of total revenue.
- The Netherlands was the second-largest market.
- The Republic of Ireland (EIRE), Germany and France were also important markets.

- The top five countries together generated about 93.9% of total revenue.

This shows that the business depended heavily on the UK. It also shows that there may be room to grow selected international markets.

8. What I Found About Cancellations

There were 9,251 cancellation records, which was 1.72% of the cleaned records.

The cancelled transactions had a total value of £893,979.73. This is equal to about 9.19% of total revenue value.

An important point is that some of the largest cancellation-value entries were not normal products. They included fees, manual entries and other financial adjustments.

Because of this, the cancellation results should not be treated as a simple product-quality score. The business would need to separate genuine customer/product cancellations from financial adjustments before making operational decisions.

9. When Sales Were Strongest

I also looked at sales by month, day and hour to understand when the business was most active.

- November 2011 was the strongest full month, with £1,456,145.80 in revenue.
- Thursday was the strongest day, with £2,108,701.53 in revenue.
- 12:00 was the strongest hour, with about £1.36 million in revenue.

December 2011 only contains data up to 9 December, so it was treated as a partial month and was not compared directly with complete months.

10. The Power BI Dashboard

I created a three-page Power BI dashboard to make the results easy to explore.

Page 1 — Executive Overview

Shows the main business figures, monthly sales, country performance and the top physical products.

Page 2 — Product & Cancellation Performance

Shows product sales, units sold, cancellation value and how cancellation value changed over time.

Page 3 — Customer & Sales Behaviour

Shows repeat and one-time customers, customer value, sales by day and sales by hour.

The dashboard is designed to give a manager a quick view of what is happening without needing to read through the raw data.

11. What the Business Could Do With These Findings

1. Focus on existing customers

Repeat customers bring in most of the identified customer revenue. The business could test loyalty rewards, personalised offers and campaigns aimed at bringing previous customers back.

2. Explore international growth

The UK is very important, but the business could look more closely at countries such as the Netherlands, Republic of Ireland, Germany and France to see where further growth is possible.

3. Understand cancellations better

The business should separate genuine product cancellations from fees and accounting adjustments. This would make it easier to identify real product or service problems.

4. Plan around busy periods

The strong Thursday and midday sales patterns could help the business plan staff, order processing and promotions around its busiest periods.

5. Review products using both sales and volume

A product can be successful because it sells many units or because it has a higher price. Looking at both measures gives a better understanding of product performance.

12. What This Project Demonstrates

This project demonstrates that I can take a large, messy dataset and turn it into a clear business story.

- I cleaned and checked the data in Excel.
- I used MySQL to answer business questions and check the numbers.
- I used Power BI to build an interactive dashboard.
- I compared results across different tools to improve confidence in the findings.
- I turned the findings into practical business recommendations.

The final project is not just a dashboard. It shows the full process from raw data, through cleaning and analysis, to business insight and recommendations.

Project repository: [online-retail-sales-customer-analytics](#)